

Ekamveer Singh

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RESEARCH INTERESTS

Reliability and robustness of multi-model LLM systems, with verification, routing, and escalation strategies that reduce hallucination and error in cascaded pipelines; efficient, on-device, privacy-preserving inference.

EDUCATION

Punjab Engineering College

B.Tech, CSE-AI (CGPA: 9.01) + Minor, Quantum Technology (CGPA: 10.00)

Chandigarh, India

Expected May 2028

PROJECTS

LLM Cascade – Minimum Sufficient Inference | *Ollama, Python, scikit-learn, Evaluation Harness* 2026

- Built a four-layer weak-to-strong ladder (gpt-oss-20b, gemma4, nemotron-3-super, nemotron-3-ultra) to study how much inference reliable task completion needs; stronger layers receive the failed candidate and execution feedback.
- On a sandboxed harness of 100 code problems, L1 alone reached 90%; execution-gated escalation reached 100% while escalating only 8% of problems (all resolved at L2), raising mean tokens only from 693 to 808 per query.
- Trained a logistic-regression router on cheap response features (no LLM judge) using TAT-QA: ~96% routing accuracy and ~76% lower latency than always using the strongest model on 50 held-out questions. Found self-reported confidence correlated negatively with correctness (−0.73). ([repo](#))

Gwen – Local Personal AI Assistant | *MLX, MLX-Whisper, SpeechBrain, Ollama, SwiftUI* Apr. 2026 – Present

- Architected a four-tier model routing system (on-device to cloud) with automatic fallback, plus a query-routing layer built to ground responses and reduce hallucination.
- Implemented speaker verification (ECAPA-TDNN) and acoustic echo cancellation (WebRTC AEC3) for full-duplex, barge-in-capable voice conversation.
- Built a fully local, privacy-first Apple Silicon pipeline (SwiftUI, FastAPI) with multimodal I/O, semantic memory, and tool use, developed independently across 15 progressive stages; v1.1 released. ([repo](#))

Automated Fungal Keratitis Screening | *Python, FastAPI, React, OpenCV, Scikit-learn* July 2026

- Led system architecture for a 3-person team building a prototype to screen fungal keratitis from confocal microscopy images, motivated by a clinical need a PGI-MER physician raised to faculty; benchmarked seven image-processing pipelines with a stacked-ensemble classifier.
- Built the LLM-based layer that turns classifier outputs and patient risk factors into clinician-readable summaries. Evaluated independently by the team; not clinically deployed.

Research ML Pipeline | *CatBoost, Scikit-learn, Pandas, OpenAlex API* 2026

- CatBoost regression pipeline predicting supercapacitor capacitance from material, synthesis, and electrolyte features, evaluated with Leave-One-Out Cross-Validation and feature analysis.

RESEARCH & PUBLICATIONS

Co-author, “Interface-Driven Electrochemical Performance of Hybrid MXenes for High-Performance Supercapacitors: Insights and Challenges” – *Current Opinion in Solid State & Materials Science*, under review. Independently authored the AI methodology section.

TECHNICAL SKILLS

Languages: Python, C++, Swift, SQL

ML / AI: PyTorch, TensorFlow, MLX, MLX-VLM, Scikit-learn, CatBoost, OpenCV

Systems: MLX Swift, SwiftUI, Apple Silicon inference, FastAPI, Ollama

Data: NumPy, Pandas, Matplotlib, SQL (MySQL, PostgreSQL, MongoDB, SQLite)

ACCOMPLISHMENTS

Finalist, Amazon ML Summer School 2026 | Published op-ed, *The Tribune* (2023) | Languages: Punjabi, English, Hindi, Norwegian